

COURSE DESCRIPTION			
Program	Polymer Technology		
Course Title	Polymer Science		
Course Code	2091	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

Polymers, the versatile macromolecules that form the backbone of countless materials in our daily lives—from plastics and fibers to rubbers and coatings. This course is designed to provide an in-depth understanding of the chemistry of polymerization and polymerization techniques, molecular weight and its determination, degradation mechanism. The concept of glass transition temperature, crystallinity and stereoregularity will help the learners to understand their effect on polymer properties. The course also emphasizes the significance of chemical modifications of polymers for advanced applications, as well as their environmental impact.

Course Objectives

1	To understand the chemistry and mechanism of polymerization and techniques used to synthesize polymers.
2	To understand the various types of polymer degradation and strategies for prevention and stabilization of polymers.
3	To develop knowledge on glass transition temperature, polymer crystallinity, and the practical methods used for polymer identification.
4	To understand various polymer reactions and modification techniques.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic Chemistry-Types of bonding, Concept of Atomic weight	2003A	Chemistry for Engineering Practices	I
Terminologies related to polymers : Monomer, Polymer, Oligomers, Functionality, Molecular weight of micromolecules and macromolecules	1091	Introduction to Polymer Science	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the chemistry of polymerization and various polymerization techniques.	Understand
CO2	Explain the concept of molecular weight and its determination.	Understand
CO3	Explain the concepts of glass transition temperature, crystallinity, and Stereo regularity.	Understand
CO4	Explain polymer degradation and polymer reactions.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
Total	12	-	-	-	-	-	-	-	-	-	12
Correlation Level	3	-	-	-	-	-	-	-	-	-	3

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-”

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total

4	0	0	0	6	60	4	40	60	100	0	0	0	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Polymerization reactions and techniques	• Polymerization ,Types of polymerization with examples-chain(addition) and step growth (condensation)polymerization. • Steps in Chain(addition) polymerization- initiation, propagation and termination, Types of initiators -free radicals, cations, or anions. • Inhibitors, Shortstops and Chain transfer agents. • Mechanism of free radical polymerization with an example.. • Ionic polymerization, Co-ordination Polymerization. • Steps in Condensation polymerization. • Polymerization techniques - Bulk, Solution , Suspension and Emulsion polymerization.	15	0
Module II	Concept and Determination of Molecular Weight	• Distinguish molecular weight of micro molecules and macromolecules. • Molecular weight of polymers - Average molecular weight concept . • Different molecular weights of polymers - Mn, Mw, Mv and Mz • Derive equations for Mn and Mw. • Polydispersity and molecular weight distribution. • Methods of determination of molecular weight of polymers.	15	0
Module III	Glass Transition Temperature,Crystallinity	• Amorphous,Crystalline and semi-crystalline polymers with examples. • Glass Transition temperature (Tg), Melting temperature (Tm), Flow temperature(Tf), Decomposition temperature(Td). • Factors affecting Tg • Determination of Tg by dilatometry. • Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. • Stereo regular polymers- Optical isomerism and Geometric isomerism • Molecular structure of IR, BR and CR	15	0
Module IV	Polymer reactions, modifications of polymer and polymer degradation.	• Polymer reactions - Differentiate reactions of polymers with simple molecules. • Effect of functional groups in polymers - Hydroxyl - Carboxylic - Aldehydic groups, Addition and Substitution reactions, Cross linking reactions. • Polymer modification- Physical and Chemical methods • Polymer Degradation- Chain and random degradation- Factors affecting polymer degradation • Types of degradation • Prevention of degradation- Antioxidant, Thermal stabilizers, U.V. Stabilizers.	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Polymerization ,Types of polymerization with examples- chain and step polymerization	Define Polymerization.	L	CO1	Remember	0.5
Polymerization ,Types of polymerization with examples- chain and step polymerization	Classify the polymerization reaction with typical examples- Chain (addition) and Step growth (condensation).	L	CO1	Understand	1.5
Polymerization ,Types of polymerization with examples- chain and step polymerization	List and explain the steps involved in addition polymerization- Initiation, Propagation, termination.	L	CO1	Understand	1

Addition Polymerization	Classify different types of addition polymerization- free radical, ionic and coordination polymerization.	L	CO1	Understand	1
Addition Polymerization	State the role of Inhibitors, Shortstops and Chain transfer agents with examples.	L	CO1	Understand	0.5
Freeradical Polymerization	Explain the mechanism of free radical polymerization with reference to initiation, propagation and termination citing an example.	L	CO1	Understand	2
Anionic Polymerization	Define Living polymers. Identify suitable monomers and catalysts for anionic polymerization.	L	CO1	Understand	0.5
Co-ordination Polymerization	Define Co-ordination Polymerization. Identify suitable monomers and catalysts for coordination polymerization.	L	CO1	Understand	0.5
Condensation Polymerization	Explain the mechanism of step growth polymerization citing an example.	L	CO1	Understand	1.5
Polymerization Techniques	List different polymerization techniques- Bulk , Solution, Suspension and Emulsion polymerization.	L	CO1	Understand	0.5
Bulk Polymerization	Explain Bulk Polymerization technique with an example.	L	CO1	Understand	1
Solution Polymerization	Explain Solution Polymerization technique with an example.	L	CO1	Understand	1.5
Suspension Polymerization	Explain Suspension Polymerization technique with an example.	L	CO1	Understand	1
Emulsion Polymerization	Explain Emulsion Polymerization technique with an example.	L	CO1	Understand	2
Module II					
Molecular weight of polymers	State and explain the concept of average molecular weight in polymers.	L	CO2	Understand	1
Molecular weight of polymers	Differentiate the molecular weight characteristics of polymers and simple molecules.	L	CO2	Understand	1
Molecular weight of polymers	List different types of polymer molecular weights .	L	CO2	Understand	1
Molecular weight of polymers	Derive equation for number average molecular weight (Mn).	L	CO2	Understand	1
Molecular weight of polymers	Derive equation for weight average molecular weight (Mw).	L	CO2	Understand	1
Molecular weight of polymers	Solve problems based on Mn and Mw.	L	CO2	Understand	1
Molecular weight of polymers	Identify techniques used to determine average molecular weight of polymers.	L	CO2	Understand	1
Molecular weight of polymers	Explain the determination of Viscosity average molecular weight by viscometry.	L	CO2	Understand	2
Molecular weight and degree of polymerisation	Explain the relationship between molecular weight and degree of polymerisation.	L	CO2	Understand	1
Molecular weight and degree of polymerisation	Explain the concept of molecular weight distribution (MWD)- broad and narrow .	L	CO2	Understand	2
Poly dispersity Index	State the significance of Polydispersity Index (PDI) of polymers.	L	CO2	Understand	1
Module III					
Glass transition temperature (Tg)-Melting temperature (Tm) -Flow temperature (Tf)	Define the terms Glass transition temperature (Tg), Melting temperature(Tm) Flow temperature(Tf) and Decomposition temperature (Td)	L	CO3	Understand	0.5
Glass transition temperature (Tg)-Melting temperature (Tm) -Flow temperature (Tf)	Explain the concept of glass transition temperature(Tg) and Identify the different factors affecting Tg of polymers.	L	CO3	Understand	2
Crystalline and amorphous polymers	Differentiate between crystalline, semi-crystalline, and amorphous polymers based on molecular arrangement.	L	CO3	Understand	2

Crystalline and amorphous polymers	List the techniques used to measure Tg and crystallinity of polymers.	L	CO3	Understand	0.5
Glass transition temperature and crystallinity	Explain the determination of Tg by dilatometry.	L	CO3	Understand	1
Glass transition temperature and crystallinity	Compare Elastomers, Plastics and Fibres in terms of Tg and crystallinity.	L	CO3	Understand	2
Polymer microstructure	Explain the microstructures based on chemical and geometrical structure citing an example.	L	CO3	Understand	2
Stereoregular polymers	Define stereo regular polymers.	L	CO3	Understand	1
Stereoregular polymers	State the importance of Z-N catalyst in stereoregularity of polymers.	L	CO3	Understand	1
Stereoregular polymers	Describe optical isomerism -isotactic, syndiotactic, and atactic.	L	CO3	Understand	1
Stereoregular polymers	Describe geometric isomerism- cis and trans isomers	L	CO3	Understand	1
Stereoregular polymers	Explain the molecular structures of IR, BR and CR.	L	CO3	Understand	1
Module IV					
Reactions of polymers and simple molecules	Differentiate between the chemical reactions of polymers and simple molecules.	L	CO4	Understand	1
Effect of functional groups in polymer Reactions	Explain the effect of hydroxyl groups, carboxylic , aldehydic groups in polymer Reactions.	L	CO4	Understand	1
Addition reaction	Explain Addition reaction in polymers	L	CO4	Understand	0.5
Substitution reaction	Explain the Substitution reaction in polymers	L	CO4	Understand	0.5
Polymer modification	State the significance of polymer modification and list physical and chemical methods used for modification.	L	CO4	Understand	0.5
Physical modification	Explain the physical modification of polymers by blending , alloying and composite preparation.	L	CO4	Understand	1
Chemical modification	Explain the various chemical methods of polymer modification by co-polymerisation.	L	CO4	Understand	1
Chemical modification	Explain the various chemical methods of polymer modification by cross-linking.	L	CO4	Understand	1.5
Chemical modification	Explain the chemical method of polymer modification by interpenetrating network structure.	L	CO4	Understand	1
Polymer Degradation	Define Polymer degradation. Classify Polymer degradation- chain end and random end degradation.	L	CO4	Remember	1
Polymer Degradation	Explain chain end and random degradation with an example.	L	CO4	Understand	1
Polymer Degradation	List the different factors affecting polymer degradation.	L	CO4	Remember	0.5
Polymer Degradation	State the advantages and disadvantages of polymer degradation,	L	CO4	Understand	0.5
Thermal degradation	Explain thermal degradation with an example.	L	CO4	Understand	0.5
Mechanical degradation	Explain mechanical degradation of polymers with an example.	L	CO4	Understand	0.5
Oxidative degradation	Explain oxidative degradation of polymers with an example	L	CO4	Remember	0.5
UV degradation	Explain the degradation of polymers by UV radiation.	L	CO4	Remember	0.5
Antidegradants and their classification	List the different antidegradants used for protecting polymers from degradation.	L	CO4	Remember	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		

1	Create a table listing monomers, mechanism, by-products, and examples for addition and step growth polymerization.	6
2	Watch a polymerization reaction video or simulation and note the differences between chain and step-growth processes.	6
3	Create a table comparing bulk, solution, suspension, emulsion polymerization for monomer type, advantages, disadvantages, and examples. Use online simulations to observe polymerization kinetics in different techniques and note differences.	6
4	Use websites like PhET, ChemCollective, or LearnChemE to simulate polymerization reactions.	6
Module II		
5	Prepare a one-page summary on importance of molecular weight in polymer processing.	6
6	Interpret molecular weight distribution curves and identify M_n , M_w , and M_z on the graph.	6
7	Explains one molecular weight determination method (osmometry, viscometry, light scattering).	6
8	Find M_n and M_w of common polymers like Polyethylene (PE), Polystyrene (PS) etc. and see how PDI affects mechanical properties.	6
Module III		
9	Make a table of common polymers, their T_g , T_m , and crystallinity.	6
10	Collect different plastic, rubber, and fiber samples from home. Classify them as amorphous or crystalline.	6
11	Use beads or colored clay to make polymer chains showing isotactic, syndiotactic, and atactic arrangements.	6
12	Comparison Chart: Elastomers, Plastics, and Fibers - Prepare a table comparing: T_g range, Degree of crystallinity, Mechanical behavior.	6
Module IV		
13	Find a commercial polymer blend or alloy and explain why it was modified.	6
14	Take two separate polymers (e.g., rubber and plastic) and imagine combining them as an IPN. Predict the changes in their properties.	6
15	Find one recent article or video on plastic pollution and degradation and summarize it. Draw a flowchart showing types of polymer degradation: Physical, chemical, biological etc.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Polymer Science	Jayadev Sreedhar, V R Gowardiker, N.V Viswanathan	New Age International Private Limited, ISBN-13 : 978-9389802580
2	Polymer Chemistry	G. S. Mishra	New Age International Pvt. Ltd. , ISBN-13:978-9386418968

REFERENCE			
Sl. No.	Title	Author	Publication
1	Textbook of Polymer Science	Fred W. Billmeyer	Wiley-Interscience; 3rd edition, ISBN-13 :978-0471031963
2	Introduction to Polymers	Robert J. Young, Peter A. Lovell	CRC Press; 3rd edition, ISBN-13 :978-0849339295
3	Introduction to Polymer Chemistry	Charles E. Carraher Jr.	CRC Press, 4th edition, ISBN-13 :978-1498737616
4	Principles of Polymerization	George Odian	John Wiley & Sons, 4th edition, ISBN-13 :978-0471274001

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://learncheme.com/screencasts/materials-science/	I

2	https://www.open.edu/openlearn/science-maths-technology/chemistry/introduction-polymers	I & II
3	https://pslc.ws/macrog/maindir.htm	All
4	https://ocw.mit.edu/courses/10-569-synthesis-of-polymers-fall-2006/	All

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60